

Rajan Kumar Gupta

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PROFILE SUMMARY

Electronics, Communication and Information Engineering graduate with hands-on experience in MERN stack, REST APIs, Supabase/PostgreSQL, and end-to-end web application development. Built production level systems including a theatre Booking System with authentication, payments, and admin dashboards. Skilled in TypeScript, React, Node.js, and IoT systems with strong problem-solving and project leadership abilities, complemented by proven mentoring experience guiding junior students in full-stack development, programming fundamentals, and hardware projects through hands-on workshops and personalized support, with a strong commitment to delivering clear, practical technical education

EDUCATION

Bachelor in Electronics, Communication and Information Engineering: Paschimanchal Campus, Tribhuvan University | Pokhara, Nepal 2021 – 2025

Higher Secondary Education: Kalika Manavgyan Secondary School | Butwal, Nepal 2018 – 2020

PUBLICATIONS

[1] A. Laudari Bhat, B. KC Khatri, G. Rijal, R. K. Gupta, and S. Adhikari, "Obstacle Detection for Visually Impaired," Proceedings of IOE Graduate Conference, Vol. 16, pp. 1913–1919, April 2025 (Published June 2026). Peer-reviewed. ISSN: 2350-8914 (Online), 2350-8906 (Print).

[IOEGC 2025 - Obstacle Detection Paper](#)

SKILLS

- **Programming Language:** JavaScript, TypeScript, Python, C/C++
- **Frontend:** React.js, Next.js, HTML, CSS, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, Django, Flask
- **Databases:** PostgreSQL, Supabase, MongoDB, MySQL, SQL Server, SQLite
- **Tools:** Git, GitHub, Postman, Render, Vercel, Netlify, Docker, AWS, Linux
- **Embedded & IoT:** Raspberry Pi, Arduino, ESP8266/ESP32, AD8232 ECG Sensor, RS485, RS232, UART
- **Others:** TensorFlow, OpenCV

Teaching and Mentoring Experience

- Technical Mentor – Paschimanchal Campus 2023 – 2024
- Mentored 15+ junior students in full-stack web development, guiding them from fundamentals to project completion
- Conducted weekly hands-on sessions covering React.js, Node.js, and database design
- Organized and led soldering and PCB design workshops for 30+ participants
- Provided one-on-one guidance for student projects and academic challenges.

WORK EXPERIENCE

Backend Intern – Digital Pathshala

May 2024 – June 2025

- Developed REST APIs for a Theatre Booking System using Node.js, Express, and Supabase.
- Implemented JWT authentication, role-based access, and Khalti payment verification.
- Built real-time seat booking, cancellation, and overlap validation logic.
- Designed optimized PostgreSQL tables and improved API response performance.
- Integrated Supabase Storage for uploading show images and user assets.

PROJECTS

Web Development Projects

- **Theatre Booking System:** Built a full-stack Theatre Booking System using React.js, TypeScript, Tailwind CSS, Node.js, Express, and PostgreSQL with Supabase. Features include JWT auth, seat booking, Khalti payment, review system, and image storage via Supabase. (<https://github.com/rajanshah23/Theatre-Booking-System>)
- **IncorpoFlow:** Company Incorporation Tool: Developed a full-stack multi-step company registration application with draft persistence and admin view. Built with React (Vite), Node.js, Express, PostgreSQL (Sequelize ORM), and Docker. Implemented instant draft saving across browser refreshes, a RESTful API with full CRUD and cascading deletes, and containerised the entire stack (PostgreSQL, backend, Nginx-served frontend) using Docker Compose for one-command setup. Includes an admin dashboard to list all companies with their shareholders.
- **Online Shopping Center:** Built a full-stack web application using MERN stack. This platform provides a seamless online shopping experience with features like product browsing, search and filter, cart management, order processing, and secure user authentication.
- **Currency Converter Web App:** Built a responsive app for real-time currency conversion using REST APIs and frontend technologies like HTML, CSS, and JavaScript.
- **Paste App:** Built a lightweight text snippet manager with React, Redux Toolkit, and LocalStorage. Features include create, edit, copy, and manage snippets with Tailwind CSS UI and React Hot Toast notifications.

Academic and Hardware Projects

- **ECG Monitoring System:** Designed and implemented ECG monitoring tool for biomedical applications with live visualization. (Paschimanchal Campus | 2023 – 2024)
- **Obstacle Detection for Visually Impaired** – Published at IOE Graduate Conference, April 2025 (<https://lnkd.in/g8TuWbnm>): Developed an assistive system using ultrasonic sensors and haptic feedback for real-time guidance. (Paschimanchal Campus | 2024 – 2025)

Leadership & Extra-Curricular Activities

- Led project teams during academic and extracurricular projects.
- Mentored juniors in full-stack development and embedded systems.
- Organized soldering and PCB design workshops.
- Participated in multiple tech events and bootcamps to stay updated with industry trends.